Physical Properties of Matter

1. **The study of production, properties of ionic state of matter in physics is called**

(a) atomic physics (b) plasma physics
(c) mechanics (d) nuclear physics
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Plasma is one of four fundamental states of matter (the other three being solid, liquid, and gas) characterized by the presence of a significant portion of charged particles in any combination of ions or electrons. It is the most abundant form of ordinary matter in the universe, mostly in stars (including the Sun), but also dominating the rarefied intracluster medium and intergalactic medium. Plasma can be artificially generated, for example, by heating a neutral gas or subjecting it to a strong electromagnetic field.

2. **Spherical form of raindrop is due to –**

(a) Density of liquid (b) Surface tension
(c) Atmospheric pressure (d) Gravitational force

39th B.P.S.C. (Pre) 1994

Ans. (b)

The shape of a drop of rain is constrained by the surface tension, which tries to give it the shape for which the surface area is minimum for the given volume. The spherical shape has the minimum surface area. That's why rain drops acquire spherical shape.

3. **The speed of light, will be minimum while passing through**

(a) water (b) vacuum
(c) air (d) glass
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

Any object or substance contains a high refractive index has the slowest speed of light. Among glass, vacuum, water and air, glass has the highest refractive index that's why the speed of light is minimum in glass.

4. **When a film of soap appears on water during the day, it shows beautiful colors. The reason of this phenomenon is-**

(a) Diffraction of light (b) Refraction of light
(c) Polarization of light (d) Interference of light
(e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

During the day, the film of soap on water appears beautiful in many colors due to interference of light rays.

5. **A piece of ice is floating in a beaker containing water up to its brim. When whole of the ice melts -**

(a) the water will spill on the floor
(b) the water level will come down in the beaker
(c) the water level will first fall and then it will spill out of the beaker
(d) the water level will not change
(e) none of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

A piece of ice floating in a beaker full of water displaces water equal to its volume, so when the ice cube melts, that water will fill the same volume and thus the water level will not change.

6. **The highest viscosity among the following is of :**

(a) water (b) air
(c) blood (d) honey
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

Viscosity is the measure of resistance of a fluid to flow. A fluid that is highly viscous has a high resistance (like having more friction) and flows slower than a low-viscosity fluid. Among the given options, honey has the highest viscosity.

7. **Name the scientist who stated that the matter can be converted into energy.**

(a) Boyle (b) Lavoisier
(c) Avogadro (d) Einstein
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (d)

Einstein Special theory of relativity explain the famous equation $E = mc^2$. This equation, which showed that energy and matter are interchangeable, provided the key to the development of atomic energy.

8. **At which temperature density of water is maximum?**

(a) 4^o C (b) 0^o C
(c) -4^o C (d) -8^o C

43rd B.P.S.C. (Pre) 1999
42nd B.P.S.C. (Pre) 1998

Ans. (a)